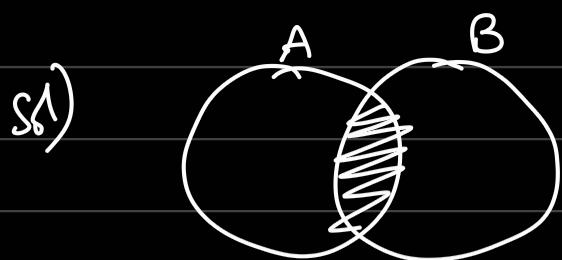


General result to remember

For any two events A, B ,

$$\max\{0, P(A) + P(B) - 1\} \leq P(A \cap B) \leq \min\{P(A), P(B)\}$$

2.14 Suppose $P(A) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$. Show that always $\frac{1}{12} \leq P(A \cap B) \leq \frac{1}{3}$.



$$\left. \begin{array}{l} A \cap B \subseteq A \\ A \cap B \subseteq B \end{array} \right\} \Rightarrow \begin{array}{l} P(A \cap B) \leq P(A) \\ P(A \cap B) \leq P(B) \end{array}$$

(By monotonicity of prob.)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0 \leq P(A \cup B) \leq 1$$

$$0 \leq P(A) + P(B) - P(A \cap B) \leq 1$$

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A) + P(B)$$

Ass: $0 \leq P(A \cap B) \leq 1$

$$\min(P(A), P(B), P(A+B), 1)$$

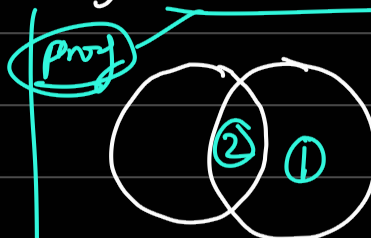
$$\max(P(A) + P(B) - 1, 0)$$

$$\leq P(A \cap B) \leq \min(P(A), P(B))$$

$$P(A) + P(B) - 1 = \frac{3}{4} + \frac{1}{3} - 1 = \frac{1}{12} - \frac{1}{4} = \frac{1}{12}$$

$$\min(1, P(A), P(B), P(A) + P(B)) = \frac{1}{3}$$

$$\therefore \frac{1}{12} \leq P(A \cap B) \leq \frac{1}{3}$$



$$B - A = B \cap \bar{A}$$

$$P((B - A) \cup (A \cap B)) = P(B - A) + P(A \cap B)$$

$$P(B) - P(A \cap B) = \underbrace{P(B - A)}_{\geq 0}$$

$$\therefore P(A \cap B) \leq P(B)$$

Let $A_1, \dots, A_n$ be given events. Show that

$$P(\cup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

where (for example) $\sum_{i < j}$ means to sum over all ordered pairs (i, j) with $i < j$.

Suppose $P(A) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$. Show that always $\frac{1}{12} \leq P(A \cap B) \leq \frac{1}{3}$.

(Subadditivity) Let $A_i \in \mathcal{A}$ be a sequence of events. Show that

$$P(\cup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i),$$

each n , and also

$$P(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i).$$

$$\sum_{i < j} P(A_i \cap A_j) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n P(A_i \cap A_j)$$

$$1 \leq i < j \leq n$$

$$i = 1 \text{ to } n-1$$

$$j = i+1 \text{ to } n$$

Just notation

changing one sigma to 2 sigma.

Inclusion–Exclusion Theorem

Let $n \geq 2$ be a positive integer, and let

$$A_1, A_2, \dots, A_n$$

be n events in the same sample space Ω . Then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n)$$

Proof — by Mathematical Induction

Let $S(n)$ denote the inclusion–exclusion formula for any n events.

1. Base case $n = 2$:

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2),$$

so $S(2)$ is true.

2. Induction hypothesis: Assume $S(k)$ is true for any k events.

3. Prove $S(k + 1)$: Take any $k + 1$ events $A_1, \dots, A_k, A_{k+1}$, and set

$$B = \bigcup_{i=1}^k A_i.$$

Then

$$P\left(\bigcup_{i=1}^{k+1} A_i\right) = P(B) + P(A_{k+1}) - P(B \cap A_{k+1}).$$

By the induction hypothesis, expand $P(B)$. Also,

$$B \cap A_{k+1} = \bigcup_{i=1}^k (A_i \cap A_{k+1}),$$

which is again a union of k events, so apply the induction hypothesis again.

Thus $P(B)$ contributes all intersection terms **not containing** A_{k+1} , while $-P(B \cap A_{k+1})$ contributes all intersection terms **containing** A_{k+1} , with the required alternating signs. Hence the inclusion–exclusion formula holds for $k + 1$ events.

Therefore,

$$S(k) \Rightarrow S(k + 1).$$

Since $S(2)$ is true, by the Principle of Mathematical Induction,

the inclusion–exclusion formula holds for every finite $n \geq 2$.

Principle of Mathematical Induction

Let $S(n)$ be a statement about integers $n \geq n_0$.

If

$S(n_0)$ is true

and

$S(k) \Rightarrow S(k + 1) \quad \forall k \geq n_0,$

then

$S(n)$ is true for all integers $n \geq n_0$.

Here, $S(k) \Rightarrow S(k + 1)$ means: assuming $S(k)$ is true, we can prove $S(k + 1)$ is true.

Principle of Mathematical Induction as a theorem

It has the form

$A \wedge B \Rightarrow C$

where:

1. A :

$S(n_0)$ is true

—the base case.

2. B :

$S(k) \Rightarrow S(k + 1) \quad \forall k \geq n_0$

—the induction step.

3. C :

$S(n)$ is true for every $n \geq n_0$.

So PMI says

$[S(n_0) \wedge (\forall k \geq n_0, S(k) \Rightarrow S(k + 1))] \Rightarrow [\forall n \geq n_0, S(n)].$

When is induction useful?

Suppose you want to prove infinitely many statements

$S(1), S(2), S(3), S(4), \dots$

Instead of proving every one separately, induction asks you to prove only:

Starting point: $S(n_0)$ is true

and

Propagation rule: $S(k) \Rightarrow S(k + 1) \quad \forall k \geq n_0.$

Then the Principle of Mathematical Induction concludes

$S(n)$ is true for every integer $n \geq n_0$.

So the main purpose is:

Induction proves infinitely many integer-indexed cases using a finite proof.

For example,

$S(n): \quad 1 + 2 + \dots + n = \frac{n(n + 1)}{2}$

is a good induction statement because we can prove a base case and show that truth for k lets us establish truth for $k + 1$.

They calling with fancy names.

$A \rightarrow$ base case

$B \rightarrow$ Induction step

1. Theorem says

$P \Rightarrow Q$

meaning: if P is true, then Q is true.

2. To use the theorem

You must first know/prove

using theorem VS proving implication
 P is true.

Then

$P, P \Rightarrow Q \Rightarrow Q.$

3. You cannot assume P just because you want to use $P \Rightarrow Q$.

4. But if your goal is to prove

$P \Rightarrow Q,$

then you may temporarily assume P and derive Q .

5. In induction

To prove

$S(k) \Rightarrow S(k + 1),$

we temporarily assume

$S(k)$ is true

and then prove

$S(k + 1)$ is true.

USE $P \Rightarrow Q$: prove P

PROVE $P \Rightarrow Q$: assume P temporarily, derive Q

Inclusion–Exclusion Theorem

Let $n \geq 2$ be a positive integer, and let

$$A_1, A_2, \dots, A_n$$

be n events in the same sample space Ω . Then

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \\ &\quad + \sum_{1 \leq i < j < r \leq n} P(A_i \cap A_j \cap A_r) - \dots \\ &\quad + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n). \end{aligned}$$

Proof by Mathematical Induction

Define the statement

$S(n)$: “The inclusion–exclusion formula is true for any n events.”

Important meaning of “any k events”

Writing

$$A_1, A_2, \dots, A_k$$

does **not** mean that we specially choose the first k events from some already fixed list.
It means:

Take any k events and label them $A_1, \dots, A_k$.

Thus, assuming $S(k)$ means that the formula may be used for **any collection of k events**.

1. Base case: $S(2)$

For any two events A_1, A_2 ,

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

Hence

$S(2)$ is true.

2. Induction hypothesis

Let $k \geq 2$. Assume that

$S(k)$ is true.

That is, assume the inclusion–exclusion formula is true for **any k events**.

We must prove $S(k+1)$.

Take any $k+1$ events

$$A_1, A_2, \dots, A_k, A_{k+1}.$$

Define

$$B = \bigcup_{i=1}^k A_i = A_1 \cup A_2 \cup \cdots \cup A_k.$$

Then

$$\bigcup_{i=1}^{k+1} A_i = B \cup A_{k+1}.$$

Using the two-event formula,

$$P\left(\bigcup_{i=1}^{k+1} A_i\right) = P(B) + P(A_{k+1}) - P(B \cap A_{k+1}). \quad (1)$$

We now expand the two terms

$$P(B) \quad \text{and} \quad P(B \cap A_{k+1}).$$

3. Expand $P(B)$

Since

$$B = A_1 \cup \cdots \cup A_k,$$

it is the union of k events.

Therefore, by the induction hypothesis $S(k)$,

$$\begin{aligned} P(B) = & \sum_{i=1}^k P(A_i) - \sum_{1 \leq i < j \leq k} P(A_i \cap A_j) \\ & + \sum_{1 \leq i < j < r \leq k} P(A_i \cap A_j \cap A_r) - \cdots \\ & + (-1)^{k+1} P(A_1 \cap A_2 \cap \cdots \cap A_k). \end{aligned} \quad (2)$$

4. Expand $P(B \cap A_{k+1})$

First use distributivity:

$$\begin{aligned} B \cap A_{k+1} &= (A_1 \cup \cdots \cup A_k) \cap A_{k+1} \\ &= (A_1 \cap A_{k+1}) \cup (A_2 \cap A_{k+1}) \cup \cdots \cup (A_k \cap A_{k+1}). \end{aligned}$$

Thus $B \cap A_{k+1}$ is again the union of exactly k events:

$$A_1 \cap A_{k+1}, A_2 \cap A_{k+1}, \dots, A_k \cap A_{k+1}.$$

Hence we may apply the induction hypothesis $S(k)$ again:

$$\begin{aligned}
P(B \cap A_{k+1}) &= \sum_{i=1}^k P(A_i \cap A_{k+1}) \\
&\quad - \sum_{1 \leq i < j \leq k} P(A_i \cap A_j \cap A_{k+1}) \\
&\quad + \sum_{1 \leq i < j < r \leq k} P(A_i \cap A_j \cap A_r \cap A_{k+1}) \\
&\quad - \dots \\
&\quad + (-1)^{k+1} P(A_1 \cap \dots \cap A_k \cap A_{k+1}).
\end{aligned} \tag{3}$$

But equation (1) contains

$$-P(B \cap A_{k+1}).$$

Therefore all signs in (3) reverse:

$$\begin{aligned}
-P(B \cap A_{k+1}) &= - \sum_{i=1}^k P(A_i \cap A_{k+1}) \\
&\quad + \sum_{1 \leq i < j \leq k} P(A_i \cap A_j \cap A_{k+1}) \\
&\quad - \sum_{1 \leq i < j < r \leq k} P(A_i \cap A_j \cap A_r \cap A_{k+1}) \\
&\quad + \dots \\
&\quad + (-1)^{k+2} P(A_1 \cap \dots \cap A_{k+1}).
\end{aligned} \tag{4}$$

5. Substitute and combine the terms

Substitute (2) and (4) into (1).

Single-event terms:

$$\sum_{i=1}^k P(A_i) + P(A_{k+1}) = \boxed{\sum_{i=1}^{k+1} P(A_i)}.$$

Pair-intersection terms:

$$- \sum_{1 \leq i < j \leq k} P(A_i \cap A_j) - \sum_{i=1}^k P(A_i \cap A_{k+1}).$$

The first sum contains all pairs **not containing** A_{k+1} , while the second contains all pairs **containing** A_{k+1} .

Therefore together they give all pairs among $A_1, \dots, A_{k+1}$:

$$\boxed{- \sum_{1 \leq i < j \leq k+1} P(A_i \cap A_j)}.$$

Triple-intersection terms:

$$\sum_{1 \leq i < j < r \leq k} P(A_i \cap A_j \cap A_r) + \sum_{1 \leq i < j \leq k} P(A_i \cap A_j \cap A_{k+1}).$$

Again, the first sum contains all triples not containing A_{k+1} , and the second contains all triples containing A_{k+1} .

Hence together,

$$\sum_{1 \leq i < j < r \leq k+1} P(A_i \cap A_j \cap A_r).$$

Why does this work for every level?

For any m -fold intersection among

$$A_1, \dots, A_k, A_{k+1},$$

exactly one of the following is true:

- (i) It does not contain A_{k+1} ,
- (ii) It contains A_{k+1} .

Equation (2) supplies all terms of type (i), and equation (4) supplies all terms of type (ii), with the correct alternating sign.

Thus no intersection term is missed and no intersection term is counted twice.

Therefore,

$$\begin{aligned} P\left(\bigcup_{i=1}^{k+1} A_i\right) &= \sum_{i=1}^{k+1} P(A_i) - \sum_{1 \leq i < j \leq k+1} P(A_i \cap A_j) \\ &\quad + \sum_{1 \leq i < j < r \leq k+1} P(A_i \cap A_j \cap A_r) - \dots \\ &\quad + (-1)^{k+2} P(A_1 \cap A_2 \cap \dots \cap A_{k+1}). \end{aligned}$$

This is exactly the inclusion–exclusion formula for $k + 1$ events.

Hence

$$S(k) \Rightarrow S(k + 1).$$

Final Conclusion

We have proved:

$$S(2) \text{ is true}$$

and

$$S(k) \Rightarrow S(k + 1) \quad \forall k \geq 2.$$

Therefore, by the **Principle of Mathematical Induction**,

$$\text{the inclusion–exclusion formula holds for every finite } n \geq 2.$$

$$\sum_{1 \leq i < j \leq K} A_i \cap A_j + \left(\sum_{i=1}^K A_i \cap A_{K+1} \right) = \sum_{1 \leq i < j \leq K+1} A_i \cap A_j$$

$$\begin{array}{l} 1 \leq i < j \\ j = K+1 \\ 1 \leq i < K+1 \\ i = 1 \text{ to } K \end{array}$$

Problem 2. Let $X \sim \text{Geometric}(p)$ and $M > 0$ be integer. Define $Z = \min(X, M)$. Compute $\mathbb{E}[Z]$.

Q2

$$Z = \min(X, M)$$

$$X \sim \text{geo}(p) \quad P(X=x) = (1-p)^{x-1} p$$

$$E[X] = 1/p$$

$$E[Z] = \sum x P(Z=x)$$

$X \geq 1$	1	2	3	4	5	...	$m-1$	m	$m+1$	$m+2$	...
Z	1	2	3	4	5	...	$m-1$	m	m	m	m

$$P(Z=1) = P(X=1) = (1-p)^{1-1} p = p$$

$$P(Z=2) = P(X=2) = (1-p)^{2-1} p = (1-p)p$$

$$P(Z=m-1) = P(X=m-1) = (1-p)^{m-2} p$$

$$P(Z=m) = P(X=m) + P(X=m+1) + \dots$$

$$= P(X \geq m)$$

$$= (1-p)^{m-1}$$

$$X \sim \text{geo}(p)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} (1-p)^{i-1} p$$

$$= \frac{p(1-p)^{k-1}}{1-(1-p)} = (1-p)^{k-1}$$

$\therefore Z$ takes values for $\{1, 2, 3, \dots, m\}$

$$E[Z] = \sum_{z=1}^m z P(Z=z) = \left(\sum_{z=1}^{m-1} z (1-p)^{z-1} p \right) + m (1-p)^{m-1}$$

Arithmetic Sum

$$\text{Let } S = (1)(1-p)^0 + 2(1-p)^1 + 3(1-p)^2 + \dots + (m-1)(1-p)^{m-2}$$

$$-(1-p)S =$$

$$pS = 1 + (1-p) + (1-p)^2 + \dots + (1-p)^{m-2} - (m-1)(1-p)^{m-1}$$

$$pS = (1) \frac{1 - (1-p)^{m-1}}{(1-(1-p))} - (m-1)(1-p)^{m-1}$$

$$pS = \left(\frac{1 - (1-p)^{m-1}}{p} \right) - (m-1)(1-p)^{m-1}$$

$$E(Z) = \left(\frac{1 - (1-p)^{m-1}}{p} \right) + (1-p)^{m-1} = \left(\frac{1 - (1-p)^m}{p} \right)$$



The answer scripts of n students are to be returned after evaluation. There are n envelopes, each bearing the name of exactly one student. Due to an error, the n answer scripts are placed in the n envelopes according to a **uniformly random permutation**, with exactly one script in each envelope.

Each envelope is then handed to the student whose name appears on it.

Find the probability that **at least one student receives their own answer script**.

*solve using
inclusion &
exclusion*

For example, for $n = 3$,

$(2, 3, 1)$

means:

- Student 1's envelope contains script 2,
- Student 2's envelope contains script 3,
- Student 3's envelope contains script 1.

Nobody receives their own script.

A **derangement** is a permutation in which **no object remains in its original position**.

Let

$D_n =$ number of derangements of n objects.

The famous formula is

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right)$$

or equivalently,

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

2. Direct answer to the question

We want

$P(\text{at least one student gets their own script})$.

Its complement is

no student gets their own script,

which is exactly a derangement.

Since all $n!$ permutations are equally likely,

$$P(\text{no one gets their own script}) = \frac{D_n}{n!}.$$

Therefore,

$$P(\text{at least one gets own script}) = 1 - \frac{D_n}{n!}$$

Using the derangement formula,

$$P(\text{at least one gets own script}) = 1 - \sum_{k=0}^n \frac{(-1)^k}{k!}$$

so

$$P(\text{at least one gets own script}) = 1 - \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right)$$

or more cleanly,

$$P(\text{at least one gets own script}) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}$$

that is,

$$1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots + \frac{(-1)^{n+1}}{n!}$$

Use the Taylor series of e^x :

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

Put $x = -1$:

$$e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}.$$

So,

$$\frac{1}{e} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots$$

Step 2: Define the events

For each

$$i \in \{1, 2, \dots, n\},$$

define

$$A_i = \{\text{student } i \text{ receives their own script}\}.$$

The event we want is

$$A_1 \cup A_2 \cup \dots \cup A_n.$$

Therefore,

$$P(\text{at least one correct}) = P\left(\bigcup_{i=1}^n A_i\right).$$

9. Now apply Inclusion-Exclusion

We want

$$P\left(\bigcup_{i=1}^n A_i\right).$$

Inclusion-exclusion says

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j)$$

Third term

Similarly,

$$\sum_{i < j < k} P(A_i \cap A_j \cap A_k) = \binom{n}{3} \frac{(n-3)!}{n!} = \frac{1}{3!}.$$

In general,

$$\binom{n}{k} \frac{(n-k)!}{n!} = \frac{1}{k!}.$$

Therefore,

$$P\left(\bigcup_{i=1}^n A_i\right) = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots + \frac{(-1)^{n+1}}{n!}$$

and hence

$$P(\text{at least one student gets their own script}) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}.$$